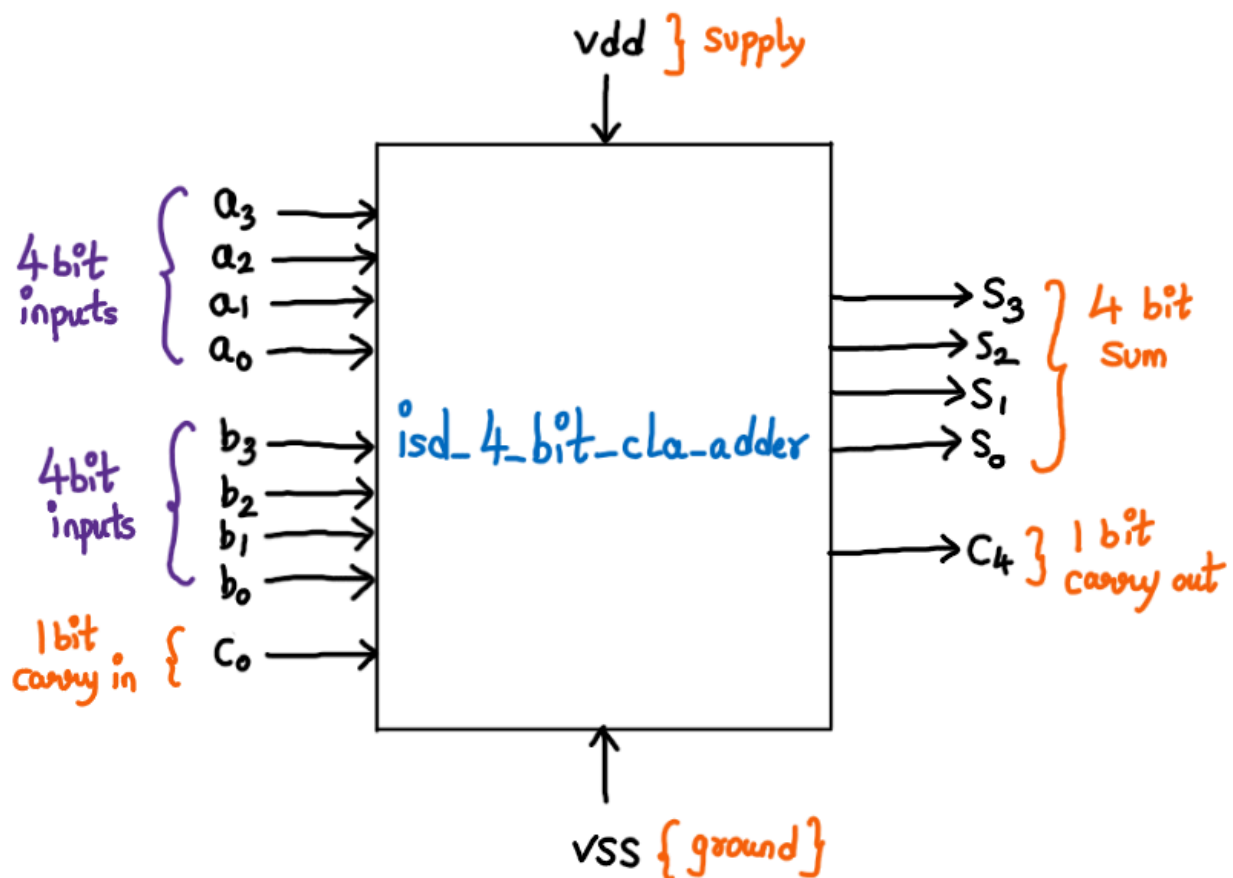


CPEG660 Project 2

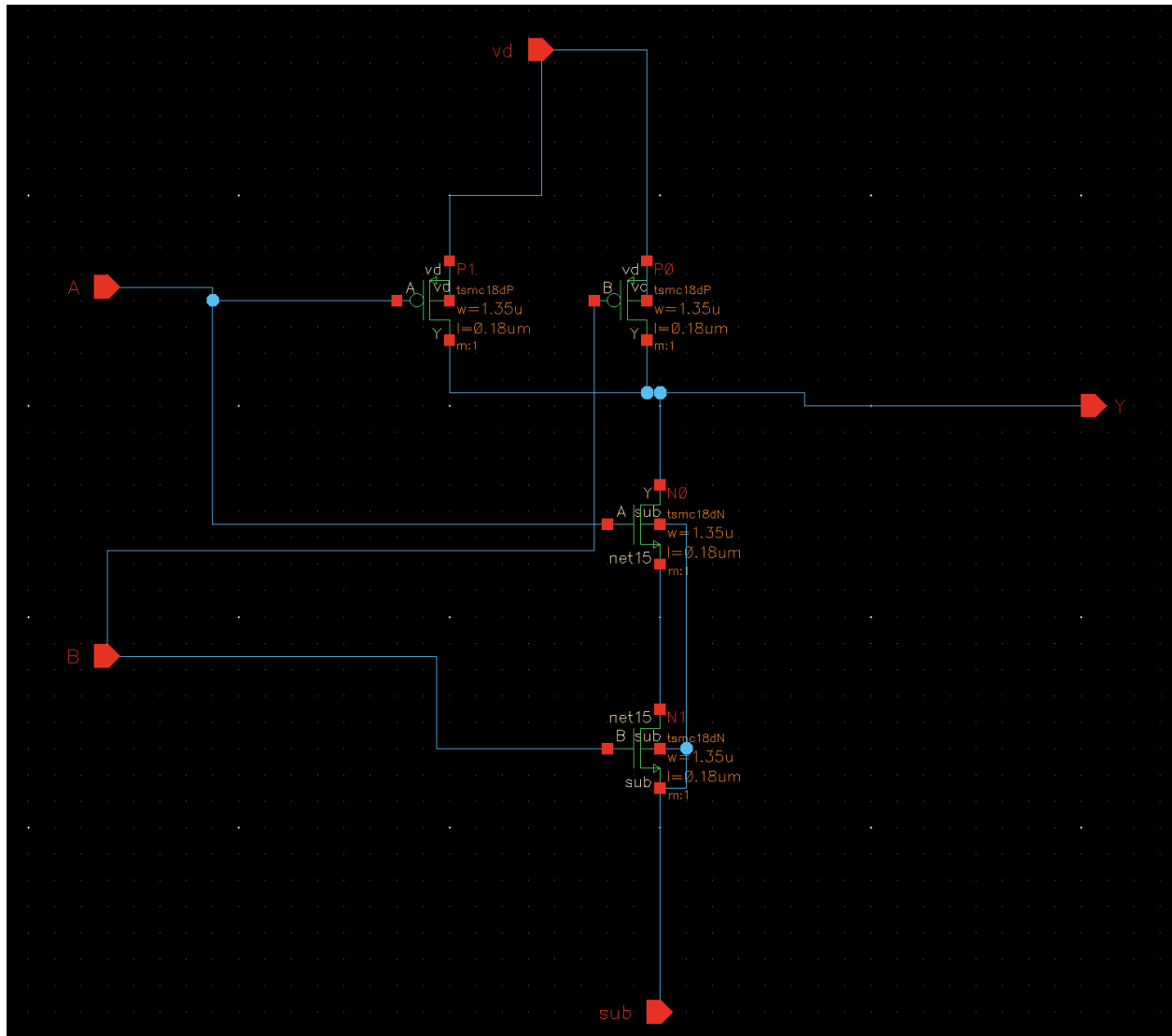
Implementation of 4 bit Carry Look Ahead Adder

A carry look-ahead adder (CLA) is a fast adder circuit that computes carry signals in advance using generate and propagate logic, so it can add multi-bit numbers without waiting for carries to ripple through each bit. CLA adder design overcomes the delay issue in conventional RCA adder by eliminating the cascading effect of the carry bits. The sum and carry terms are processed at once in the CLA adder.

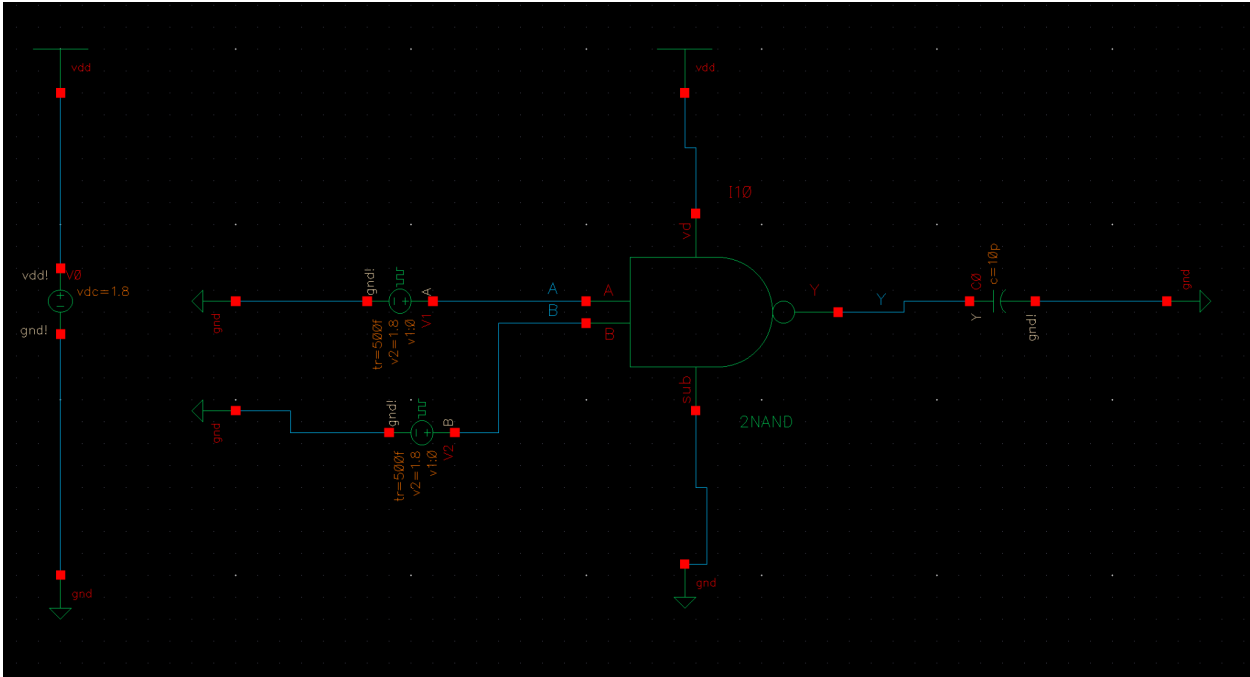


NAND gate

schematic

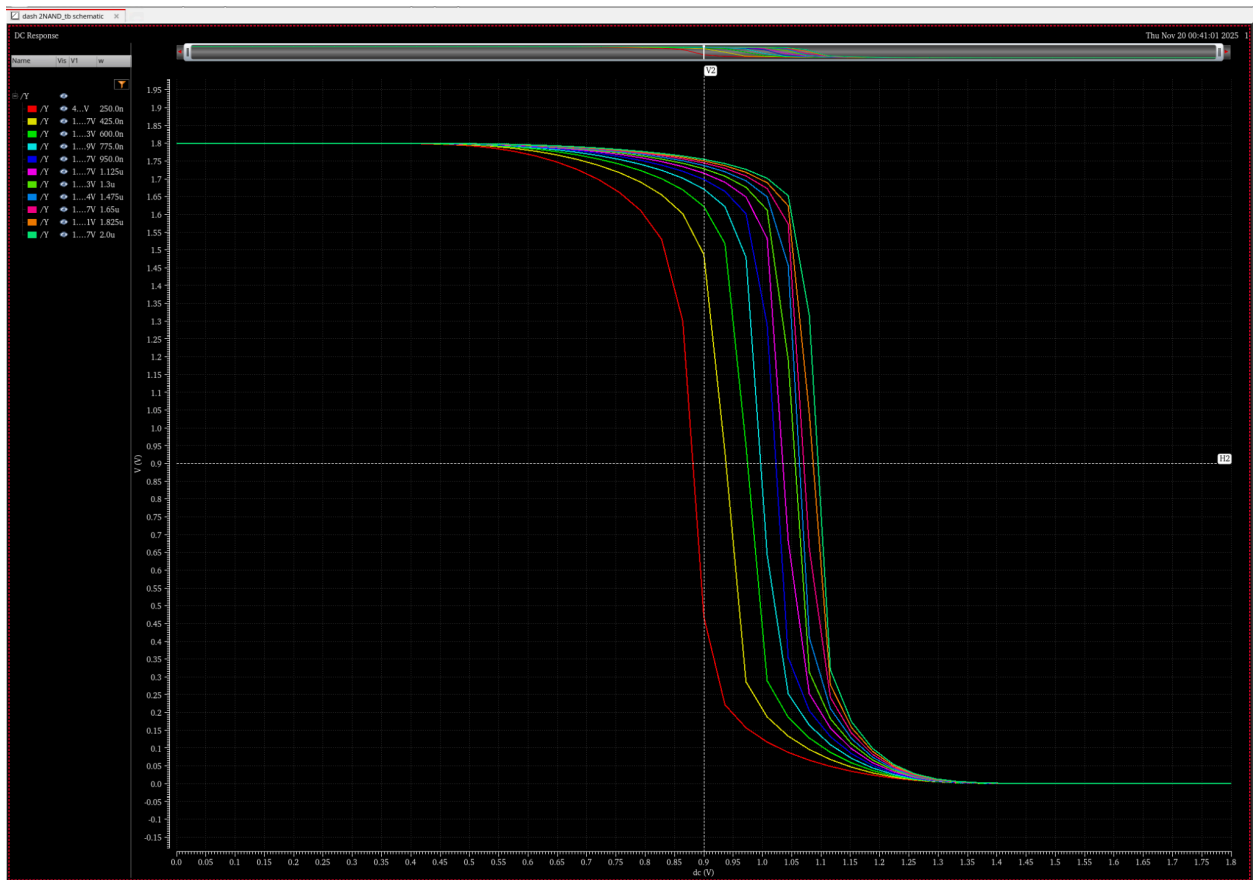
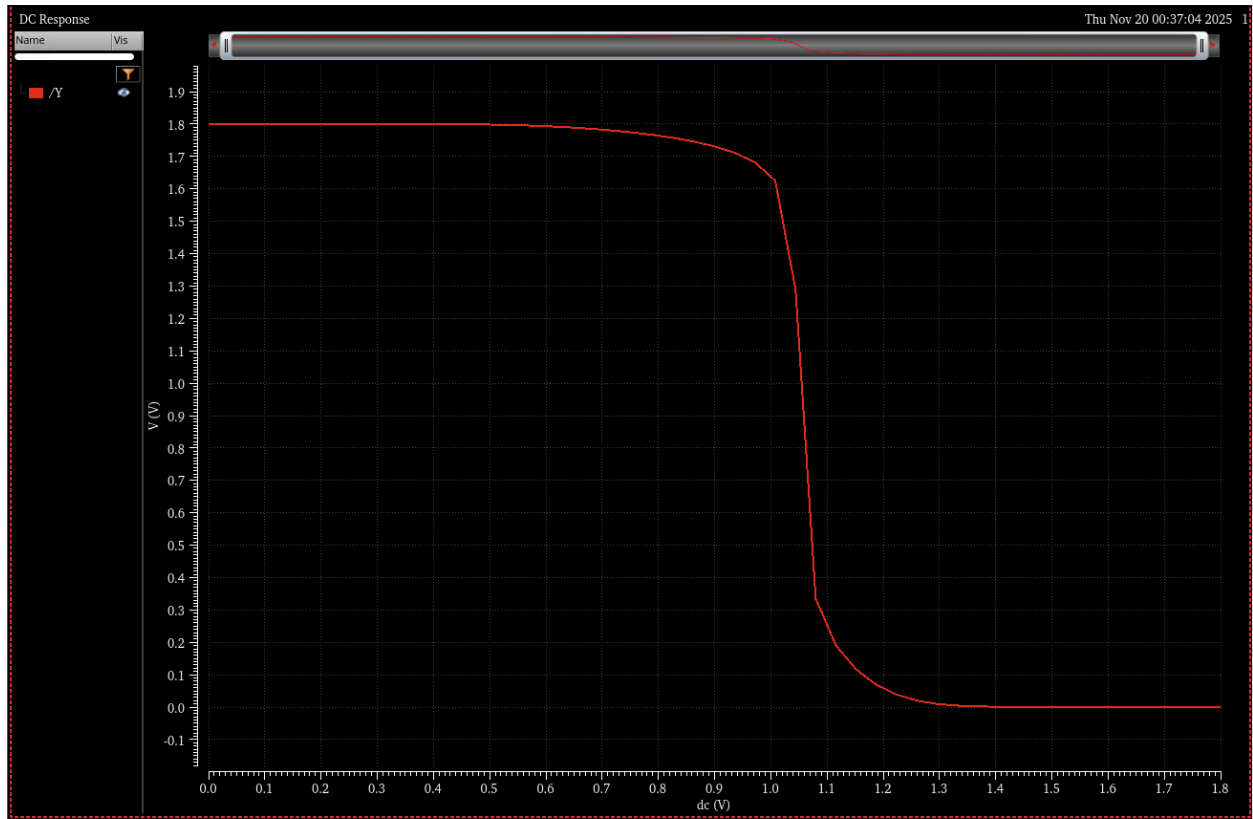


Test Bench



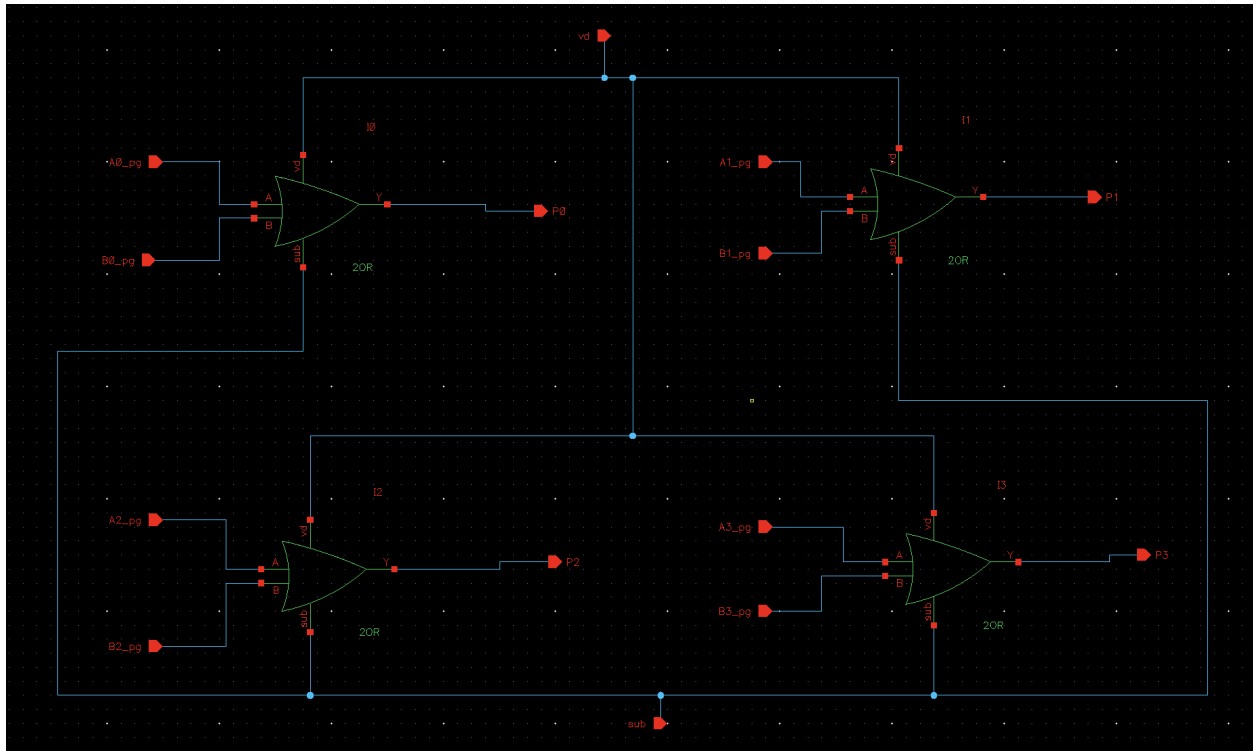
Analysis





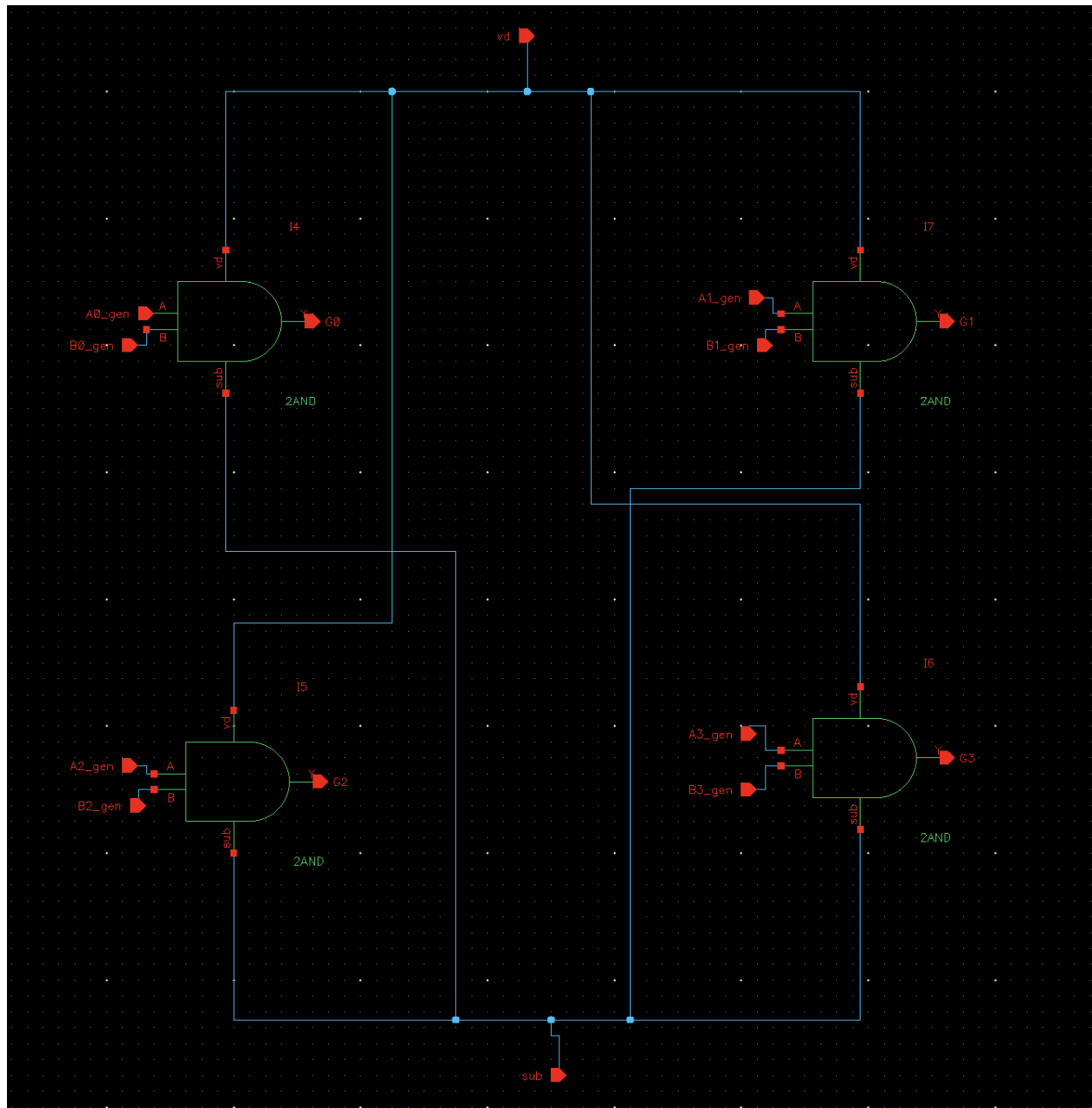
Propagate Circuit

- $P_i = A_i + B_i$
- Made using OR gates
- Consists of 24 transistor



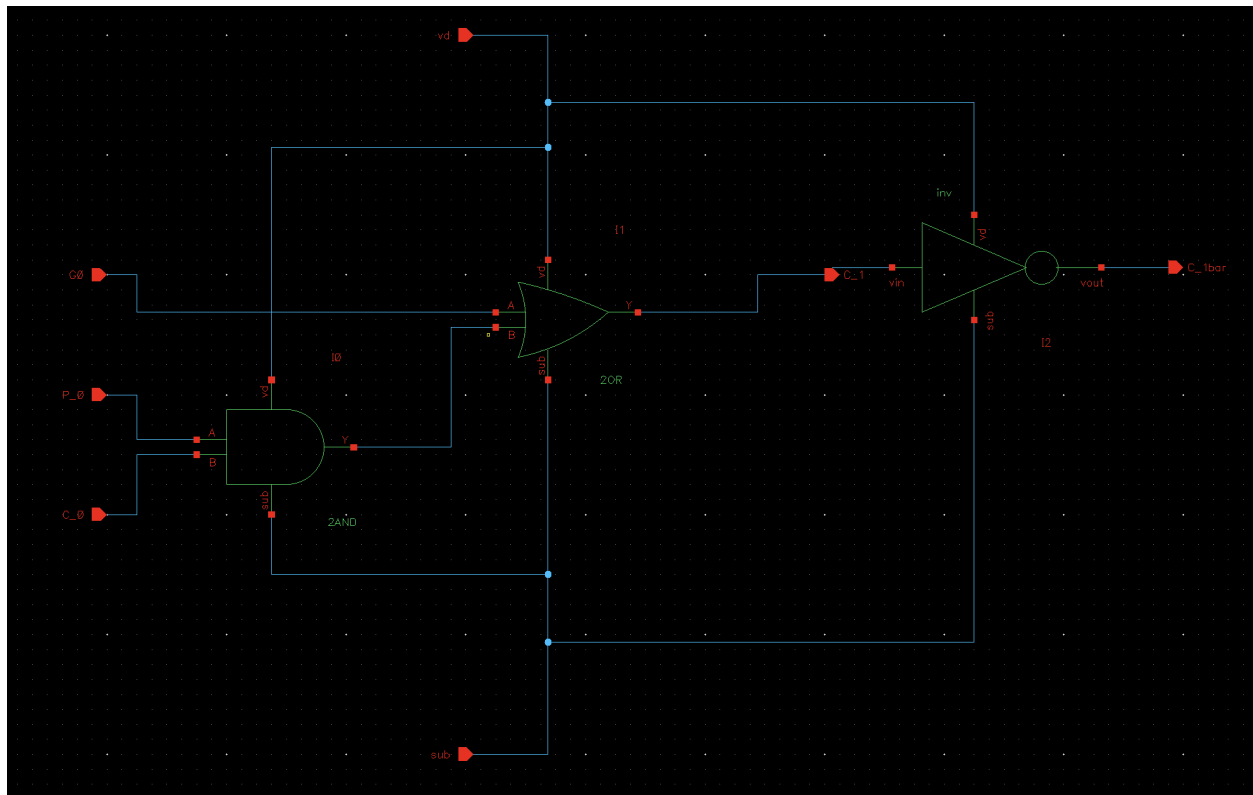
Generate Circuit

- $G_i = A_i B_i$
- Made using AND gates
- Consists of 24 transistors



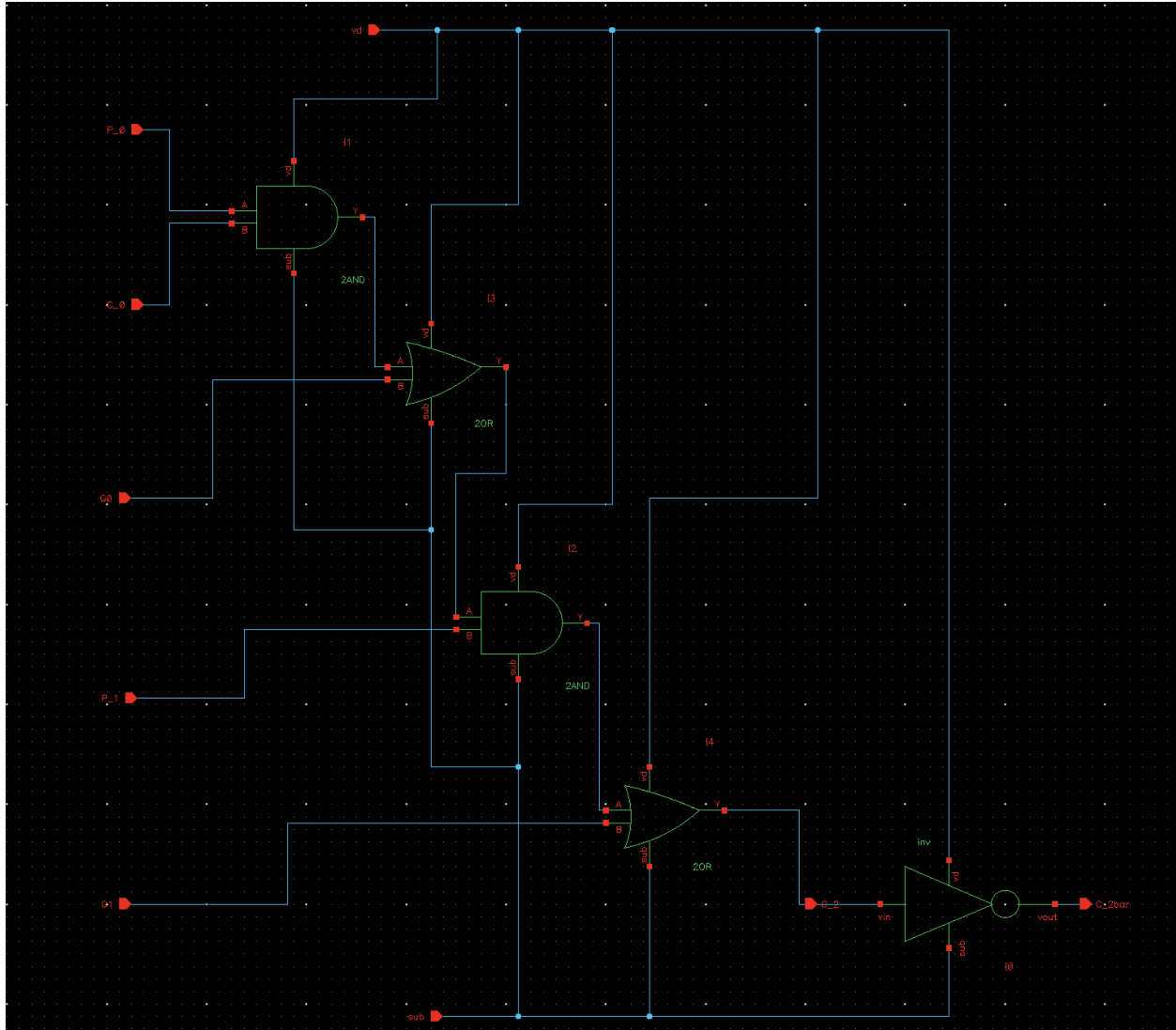
Carry 1 block

- $C1 = G0 + P0C0$
- Consists of 8 transistors



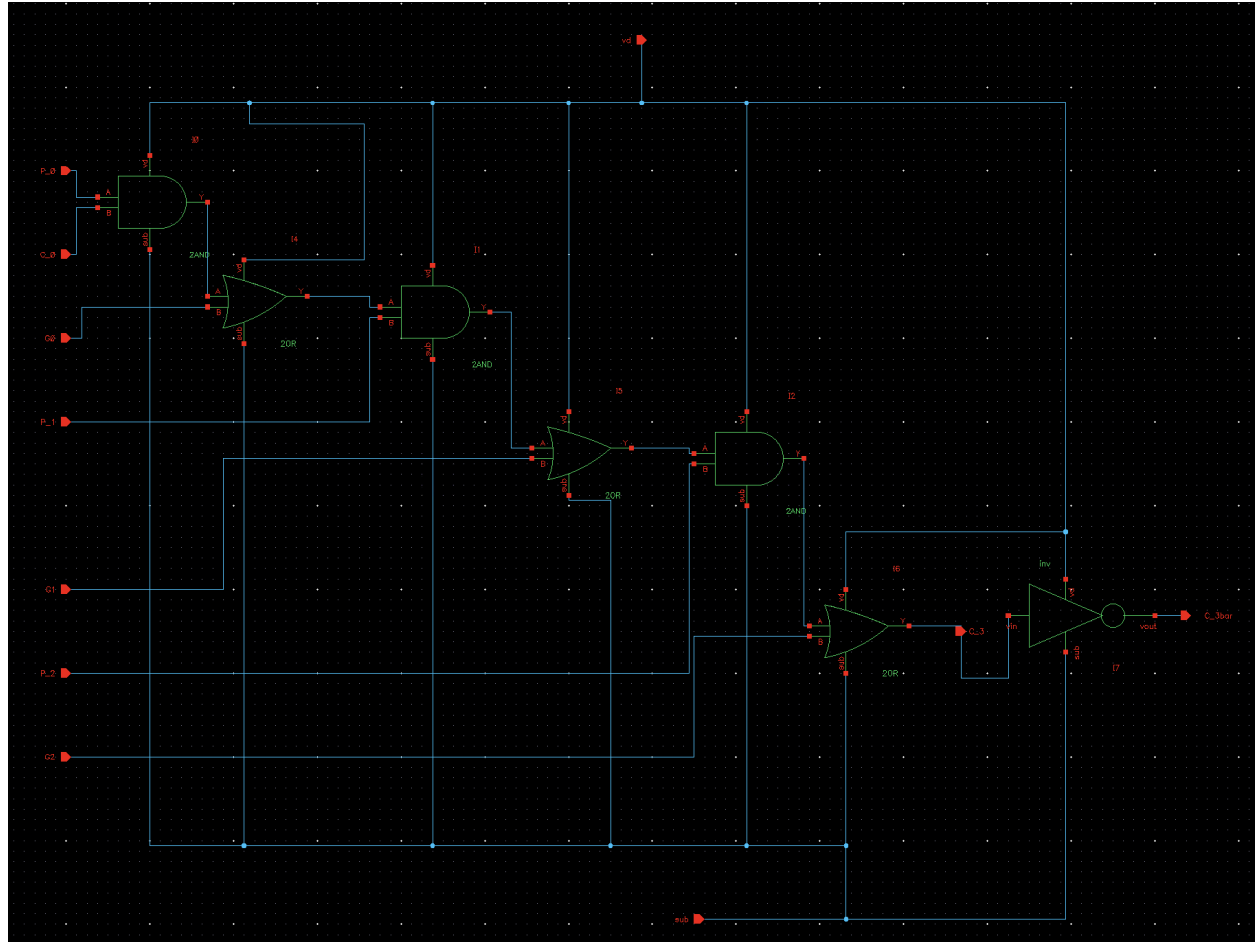
Carry 2 Block

- $C2 = G1 + P1(G0 + P0C0)$
- Consists of 12 transistors



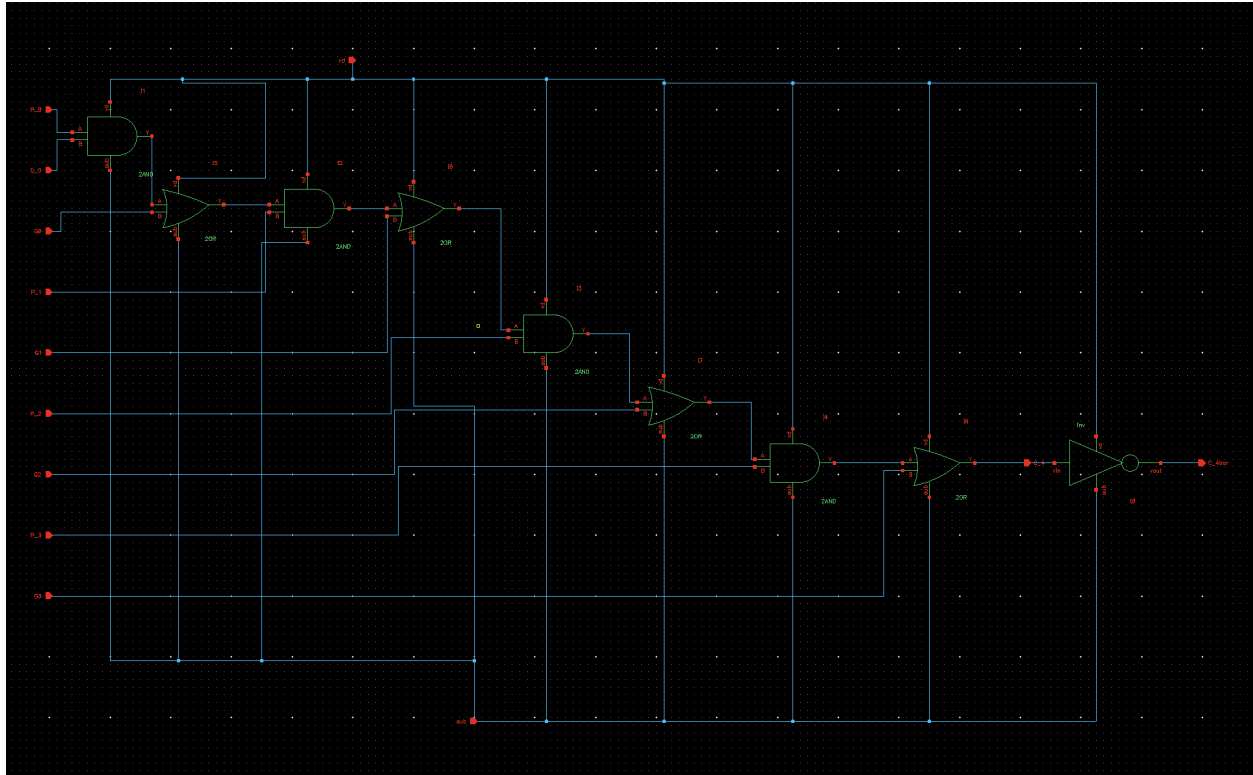
Carry 3 Block

- $C3 = G2 + P2(G1 + P1(G0 + P0C0))$
- Consists of 16 transistors



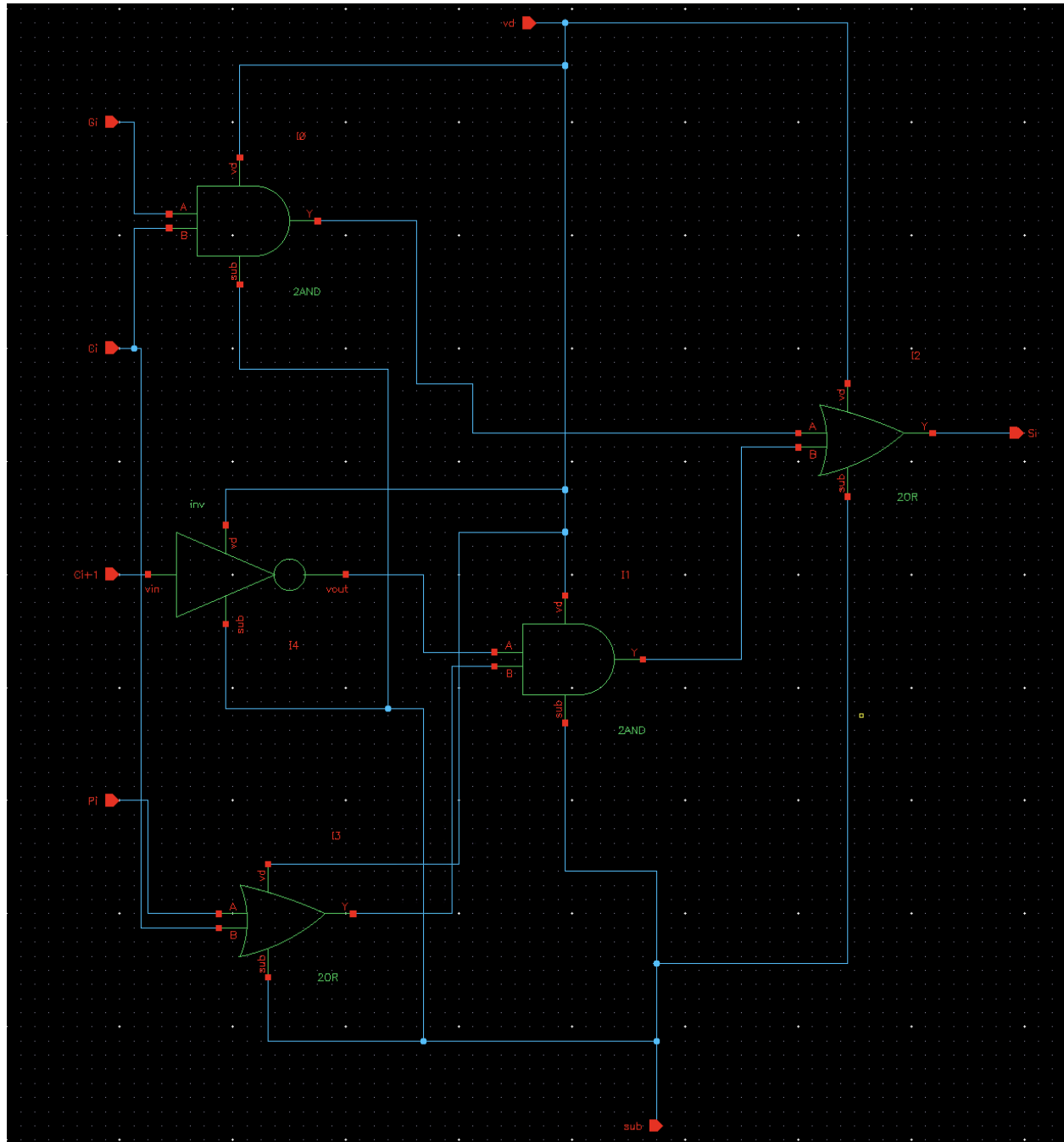
Carry 4 Block

- $C4 = G3 + P3(G2 + P2(G1 + P1(G0 + P0C0)))$
- Consists of 20 transistors



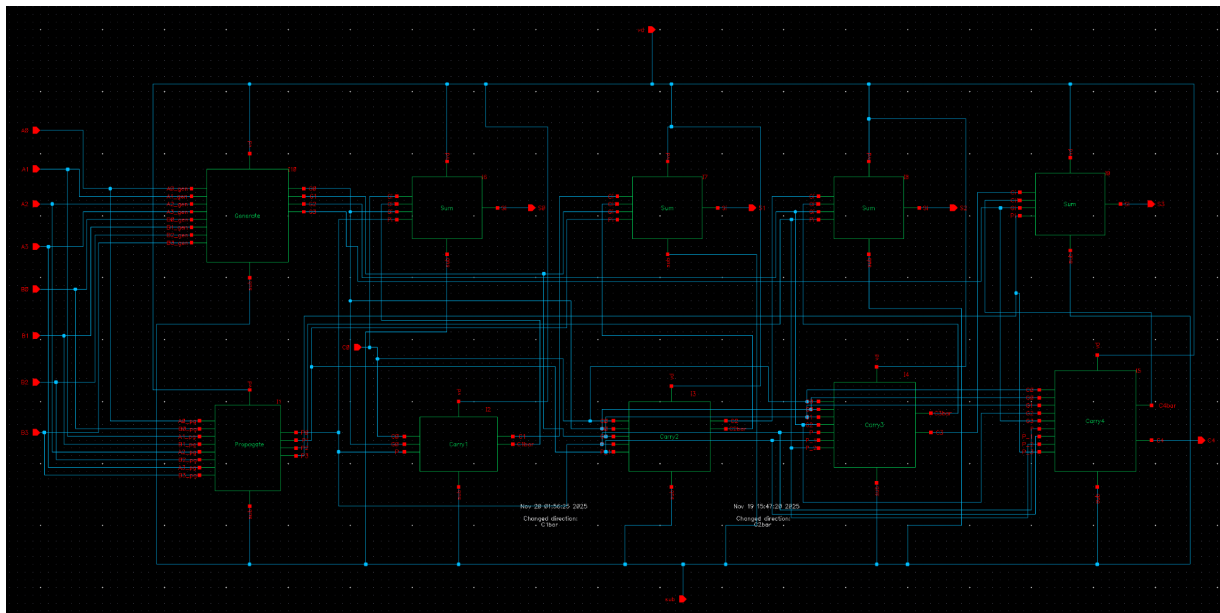
Sum Circuit

- $S_i = G_i C_i + (C_i + 1) \bar{P}_i (P_i + C_i)$
- Consists of 12 transistors

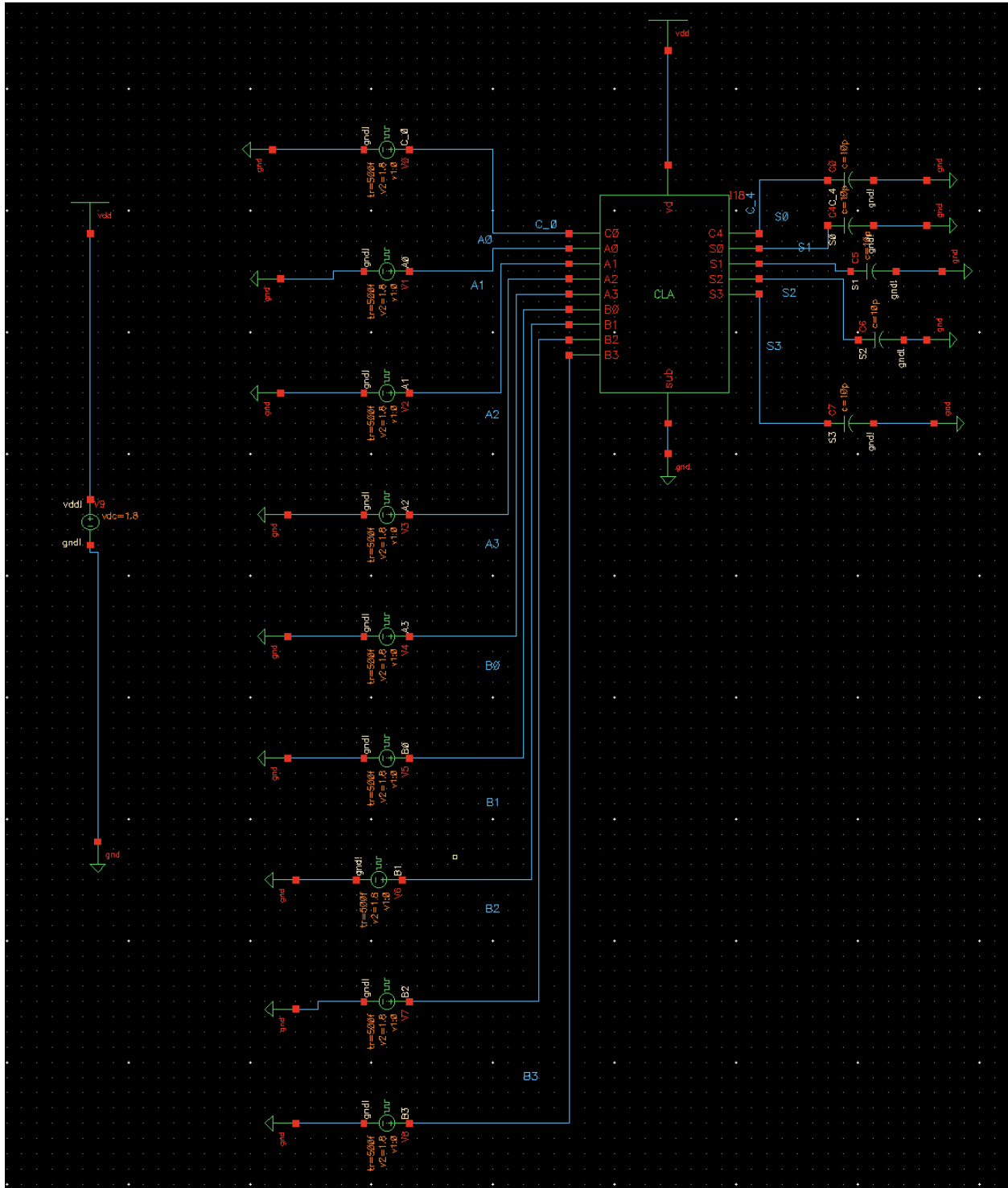


4 bit CLA adder Schematic

- All 8 inputs A0-A3 and B0-B3 are applied to generate and propagate blocks.
- The outputs from generate and propagate blocks are G0-G3 and P0-P3.
- With the help of propagate and generate terms, sum and carry bits are evaluated in a faster way.
- Four sum circuit blocks and carry 1 to carry 4 circuit blocks are interconnected together to achieve a 4 bit CLA adder.
- Total transistor count for the 4 bit CLA adder design is 152.



Testbench



Wave form

